

California Housing Prices – EDA & Linear Regression Summary

Dataset Overview

The dataset contains housing-related features such as location, population statistics, and median income, with the target variable being median house value.

Exploratory Data Analysis Findings

- Missing values are present only in *total_bedrooms*.
- Most numerical features are right-skewed (population, rooms, income).
- Median house value is capped at \$500,000, affecting predictions.
- Strongest correlation with target: *median_income*.

Modeling Approach

A Linear Regression model was trained using a preprocessing pipeline including median imputation for numerical features and one-hot encoding for categorical features.

Model Performance

- MAE $\approx$ 50K
- RMSE $\approx$ 70K
- $R^2 \approx 0.6$

Residual analysis shows heteroscedasticity and underprediction for high-value houses.

Key Conclusions

Linear Regression provides a reasonable baseline but struggles with non-linear patterns. Further improvements can be achieved through feature engineering and non-linear models.